A Compact Autonomous Flapping-Wing Aerial Vehicle: Design, Modeling, and Vision-Based Control for Narrow Gap Traversal

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***Abstract*—Aggressive autonomous flight through narrow gaps presents a critical challenge in drone racing. As flapping-wing aerial vehicles (FWAVs) evolving toward autonomous navigation, avian-inspired FWAVs are significantly hindered by their single-modal flight dynamics and nonlinear underactuated characteristics, leading to the largely unexplored area of narrow gap traversal. To address this gap, this article introduces *SparrowHawk*, a compact autonomous flapping-wing system optimized for aerodynamic efficiency and structural rigidity. Its design enables a favorable payload-to-weight ratio, accommodating onboard sensing for external-positioning-free localization and maneuverable forward flight. By leveraging the cycle-averaged method for modeling, the system is simplified to a nonlinear time-invariant model, with kinematic and dynamic equations derived in a unified state-space formulation. For robust real-time gate detection during flight, an adaptive Hough transform-based visual algorithm is proposed, yielding pixel-space geometric parameters of gaps without any prior knowledge. The servo control framework employs a closed-loop dual-loop strategy, integrating a TECS-based altitude controller with a cascaded PID attitude controller to ensure precise position and orientation control during autonomous gap traversal. Indoor and outdoor flight experiments demonstrate that *SparrowHawk* executes tight-radius banked turns and altitude control with a mean error of 0.345 m. The robot successfully traverses an 80 cm-diameter circular gap at speeds up to 5.6 m/s with a 60 cm wingspan, signifying the first achievement for FWAVs. Through multiple repeated trials, it achieved success rates of 80% under normal lighting and 60% under low-light conditions, significantly outperforming a human pilot, the conventional unmodified Hough circle detector, and the deep learning–based YOLOv11 algorithm. This work advances perception-driven control methodologies for FWAVs, paving the way for agile autonomous navigation in confined environments.**

***Index Terms*—Autonomous navigation, biologically-inspired robots, dynamics modeling, flapping-wing aerial vehicle (FWAV), mechanical design, vision-based control.**

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I. Introduction

BIO-INSPIRED flapping-wing aerial vehicles (FWAVs) have emerged as a promising paradigm in the field of aerial robotics. These robots mimic the flight mechanisms of birds and insects by producing aerodynamic lift and thrust through periodic wing flapping, coupled with passive wing deformation and torsional motion [1]. Benefiting from their unique biomimetic features, such as multimodal transitions [2], agile maneuverability [3], perching and grasping capabilities [4], [5], and superior aerodynamic efficiency at low Reynolds numbers [6], [7], FWAVs exhibit considerable potential for applications in industrial inspection [6], [8], emergency response and search-and-rescue [7], autonomous operations in confined or complex industrial spaces [9], [10].

With ongoing advancements in unsteady aerodynamics and overall design, FWAVs have steadily matured and are generally categorized into two types: micro-scale insect-inspired tailless designs, exemplified by the DelFly Explorer & Nimble [11], [12], KUBeetle-S [13], [14], NUS-RoboticBird [15], and medium-to-large-scale avian-inspired platforms, including Falcon [1], Dove [6], E-FLAP [16] HIT-Hawk & Phoenix [17], USTB-Hawk [18], and Cloud-Owl [19]. Alongside the evolution of these platforms, the demand for autonomous navigation in constrained environments has increasingly emerged [10], [20]. In nature, birds such as sparrowhawks and falcons showcase extraordinary agile maneuvers to navigate through narrow spaces, evade obstacles, and capture prey [1], [21]. However, compared with their biological counterparts and other types of MAVs, current FWAVs still face significant challenges and unresolved research gaps in achieving fully autonomous navigation. A typical application is the gap traversal problem, which demands precise whole-body perception and control when navigating at

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high speeds through confined environments. These challenges can be summarized as follows.

1) *High-payload flight challenges:* The inherently limited wing loading capacity and strict power constraints severely restrict the selection of onboard perception sensors and avionics systems.
2) *Onboard perception challenges:* Significant periodic oscillations and mechanical vibrations induced by the reciprocating flapping motion introduce additional measurement noise, which manifests as camera motion blur and IMU bias accumulation.
3) *Perception-driven servo control challenges:* The nonholonomic underactuation and strong aerodynamic coupling complicate precise dynamic modeling. Unlike hover-capable vehicles, which are capable of multiple flight modals, the tailed design of this FWAV restricts it to high-speed forward flight, posing challenges for precise whole-body autonomous vision-based control in confined environments.

To address the aforementioned challenges, this article presents a compact and autonomous avian-inspired FWAV named after the Sparrowhawk, one of the smallest-winged hawks in nature. We provide a comprehensive overview of the design, modeling, development, and vision-based control methods. The main contributions of this work can be summarized in three main aspects. First, a novel compact FWAV platform is developed, integrating aerodynamic layout optimization to achieve an exceptional payload ratio exceeding 50% while maintaining high maneuverability. The platform incorporates a lightweight, high-resolution monocular vision system, a multi-redundant IMU flight controller, and an optical flow module for autonomous self-localization, with image processing for remote control performed at the ground station. Furthermore, a cycle-averaged formulation is employed to systematically derive a physically consistent state-space representation, providing a solid foundation for vision-based control. Second, a real-time, efficient sliding window-based visual gate detection algorithm that incorporates an adaptive mechanism and leverages the Hough transform is proposed. The algorithm ensures robust and accurate gate recognition under rapid motion and scale variations, without requiring any prior knowledge. The servo controller implemented on PX4 firmware with offboard control is integrated into a dual-loop framework that employs vision, IMU, and optical flow fusion for 6-DoF state estimation without external localization, facilitating agile and reliable gate traversal. Finally, extensive experiments were conducted to validate the effectiveness of the overall design and the vision-guided controller, including outdoor maneuvering flights, altitude-hold flights, comparative evaluation of visual detection algorithms on two distinct datasets, and indoor circular gate traversal tests. To the best of our knowledge, this is the first practical implementation of a small-scale flapping-wing robot that integrates real-time vision-based perception with offboard flight control. It successfully demonstrates the autonomous traversal of an 80 cm narrow gate by a robot with a 60 cm wingspan, achieving a maximum speed of 5.6 m/s while maintaining stable forward flight and exceptional maneuverability. Multiple repeated experiments measuring the success rate under varying illumination confirmed the system's stability and scalability. This work bridges the gap between autonomous perception and offboard control for FWAVs, pioneering high-speed narrow obstacle avoidance capabilities comparable to the natural flight abilities of avian species.

The rest of this article is organized as follows. Section II reviews the research progress in MAV aggressive gate traversal and autonomous perception and avoidance control methods for FWAVs. Section III introduces the design and avionics of the Sparrowhawk platform. Section IV presents the kinematic and dynamic modeling of the Sparrowhawk, along with the corresponding state-space equations. Section V provides a detailed explanation of the visual-guided controller design. In Section VI, the real-world flight, the comparison of the gate detection algorithm, and narrow-gap traversal experiments are conducted. Section VII presents the comparison and validation with other similar FWAV platforms, as well as the static flapping endurance tests. Finally, Section VIII concludes this article.

II. Related Works

Autonomous aggressive flight through narrow gaps has been extensively explored in quadrotors and is regarded as one of the most challenging problems [22]. This task demands whole-body perception and control, along with the precise and rapid coordination of the vehicle's position and attitude within a limited response time. The traditional decoupled approach follows a localization-planning-control pipeline. It leverages nonlinear optimization, employing the differential flatness of polynomials and B-splines to generate dynamically feasible smooth trajectories, with model predictive control (MPC) or differential flatness control serving as the controllers [21], [23], [24], [25]. Conversely, an alternative approach utilizes a learning-based end-to-end policy that directly generates low-dimensional control commands from high-dimensional observations, enabling the establishment of a Sim2Real transfer from simulation to the real world [22], [26], [27]. The Swift system [28], [29], trained in simulation through deep reinforcement learning, integrates feedback neural networks into the control policy and uses nonparametric empirical noise models to address Sim2Real transfer discrepancies. This work marks a milestone in which learning-based methods have surpassed human pilots in drone racing.

Inspired by the aggressive traversal through narrow gaps exhibited by quadrotors and drone racing, performing similar tasks with FWAV platforms remains an unexplored research area. Early work related to perception-driven avoidance control primarily focused on tailless micro insect-inspired robots. DelFly Explorer and Nimble pioneer the application of the appearance variation cues [30], "Droplet" strategy [12], monocular snake-gate detection [10] and stereo visual semi-global optimization LongSeq algorithms [31]. While the Delfly Explorer demonstrates the ability to explore multiple rooms and traverse gates, it is limited by relatively slow obstacle avoidance flight speeds and its reliance on external motion capture systems for localization. Within the domain of medium and large-scale FWAVs, the FWAF-VID [32], and GRIFFEN datasets [33] represent significant advancements in onboard self-localization and event-camera robust perception capabilities. Ryu et al. [20] propose a vision-based target tracking method that employs color segmentation for visual detection, recognizing the color block, and directly converting the error into pitch and yaw voltage control commands. However, the direct error mapping without

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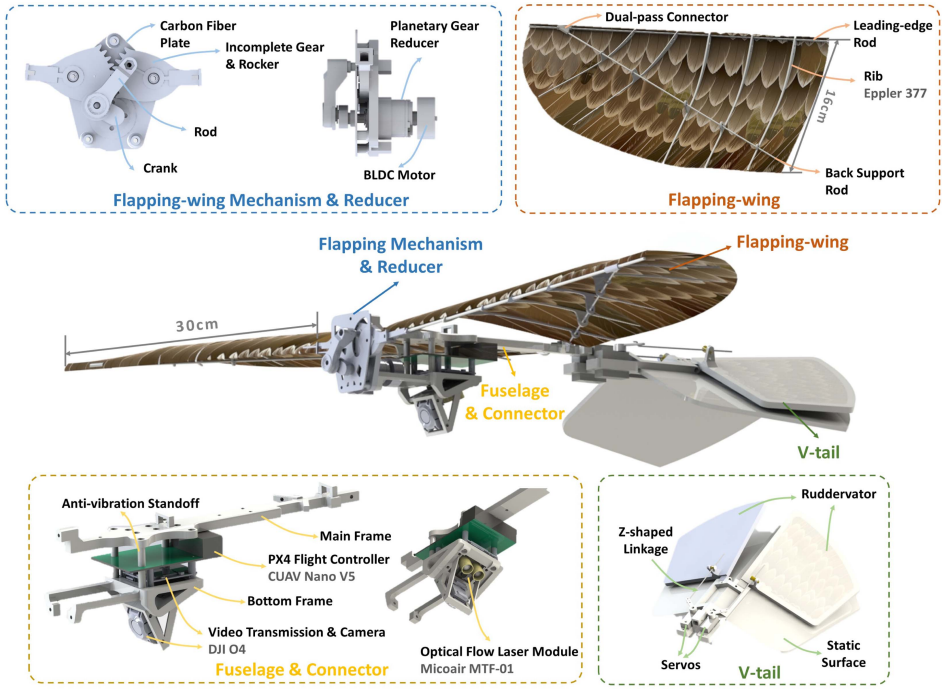


Fig. 1. Assembly and modular structural design diagram.

a closed-loop feedback controller cannot guarantee the stability and accuracy of system. Eguiluz et al. [34] propose an onboard event camera-based closed-loop control method that extracts triangular line features for visual servoing, facilitating autonomous target-oriented maneuvering, but without achieving narrow gate traversal. Learning-based approaches have gradually emerged in FWAV autonomous flight research. Lee et al. [35] support a learning-based architecture with MPC control for path tracking, but it lacks real-world experimental validation. Chen et al. [36] propose a reinforcement learning-based event-triggered planner for autonomous obstacle avoidance. However, the intricate and unsteady aerodynamics of FWAVs continue to pose difficulties in high-fidelity modeling, dynamic parameter identification, and Sim2Real transfer accuracy.

In response, this article presents a flapping-wing robot platform Sparrowhawk with an optimized payload-to-weight ratio for onboard self-localization. A cycle-averaged kinematic and dynamic model is derived to support control strategies, while a real-time vision-based circular gate detection method ensures efficient gate recognition. Finally, a dual-loop closed-loop control method is implemented, enabling the first aggressive maneuver through a narrow circular gate.

III. Design and Avionics of Sparrowhawk

A. Mechanical Design

The mechanical design of the platform consists of four primary modules: the flapping mechanism and reducer, flapping wings, fuselage, and V-tail, as illustrated in Fig. 1. The flapping mechanism utilizes a compact four-bar linkage, with the rocker directly functioning as a reciprocating incomplete gear. This integration not only minimizes the number of components but also facilitates symmetric wing actuation by mechanically driving the opposing gear without the need for additional linkages. A fuselage-axis-aligned planetary reducer, paired with the brushless direct current motor (BLDC) motor, provides the required torque within a compact configuration. The flapping wings are designed based on the high-speed wing morphology of Sparrowhawk, which features greater stiffness on the inner surface for lift production and enhanced flexibility on the outer surface for passive deformation. The lift coefficient was determined from design parameters, including maximum takeoff weight, cruising speed, and wing area, with Reynolds numbers ranging from 1.0e5 to 1.5e5. The Eppler 377 airfoil was chosen for the outer wing rib profile after comparisons with the airfoil library. In addition, weight reduction measures were implemented for the inner wing rib profile, and holes were added to secure the back support rod. The wings exhibit asymmetric stroke amplitudes, with an upstroke angle of 65° and a downstroke angle of 20°, generating rolling moments that enhance lateral agility and roll responsiveness during turns.

The fuselage is designed for the integration of the flapping mechanism, V-tail, flight controller, and video transmission module. To facilitate modular assembly and minimize structural weight, both the main and bottom fuselage components are manufactured integrally using PLC-based 3-D printing, and further reinforced with carbon sheets to improve structural stiffness and strength. The PX4 flight controller and DJI O4 video transmission unit are mounted using isolating standoffs to mitigate mechanical vibrations caused by oscillatory motion. The bottle fuselage incorporates a monocular camera and a MicoAir MTF-01 optical flow and laser sensor. To ensure the camera maintains a forward-facing orientation and the optical flow sensor remains in a downward-facing configuration during forward flight, where the vehicle typically maintains a pitch angle of approximately 25°, the mounting bracket is deliberately inclined to compensate for this flight attitude. To enhance the bio-inspired fidelity of the platform and mitigate the inherent maneuverability limitations due to the absence of conventional ailerons on wings, a V-tail

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TABLE I
MAIN SPECIFICATION

Performance indicators	Value
Total weight (g)	215
Maximum payload capacity (g)	75
Wingspan (cm)	60
Wing loading (kg/m^2)	3.16
Chord length (cm)	16
Aspect Ratio	5.29
Flapping frequency range (Hz)	0–8.0
Flight speed range (m/s)	4.0–10.0
Flapping amplitude (degree)	85(+60, −25)
Maximum endurance (min)	20

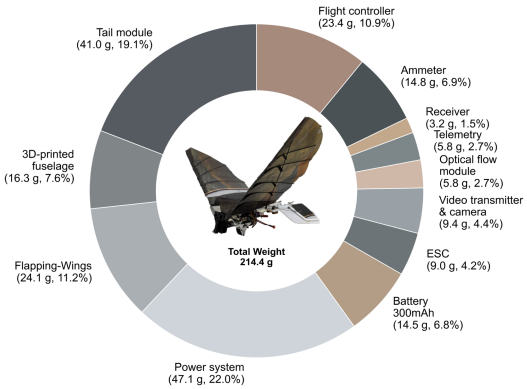


Fig. 2. Mass distribution of sparrowhawk components and sensors.

configuration is employed for the tail assembly. The V-tail is actuated by two independent servos that differentially control aerodynamic flaps to generate asymmetric lift for roll control, while symmetric deflection produces pitch moments for attitude regulation. The V-tail is configured with an inclination angle of 65°, which facilitates the generation of roll moments during turning maneuvers. To prevent potential dynamic instability, such as spin entry induced by fully actuated control surfaces under aggressive flight conditions, and to enhance longitudinal static stability, fixed aerodynamic surfaces made of KT board are mounted beneath the control flaps. The main specification of the platform are shown in Table I. The mass distribution of Sparrowhawk is depicted in Fig. 2.

B. Avionics and Sensors Integration

Due to the strict constraints on payload and power consumption of Sparrowhawk, it is impractical to equip it with a minimal onboard computer running a Linux system for real-time image processing. Consequently, to ensure stable flight performance, the avionics and sensor integration utilize a ground-based processing scheme, in which both video and data streams are transmitted back to the ground station. The onboard electronic system is powered by a 2S 300-mAh LiPo battery connected to the flight controller via a DC–DC converter with an ammeter. The flight controller utilizes the CUAV Nano V5, which is flashed with PX4 firmware version 1.16.0. It is powered by an STM32F765 MCU and integrates a triple-redundant IMU, an electronic compass, and a barometer. This configuration enables accurate attitude estimation and provides reliable control signal output. An RC receiver communicates with the controller through the s.Bus protocol. The Wi-Fi telemetry module is based

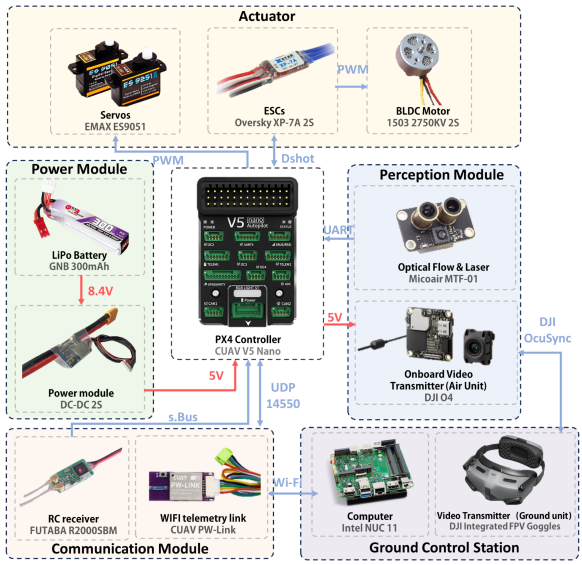


Fig. 3. Layout of the avionics system and onboard sensors.

on an ESP8266 MCU, which establishes a low-cost UDP 14 550 broadcast link between the flight controller and a ground-based Intel NUC computer, enabling bi-directional transmission of control commands. The Ground Control Station is composed of a DJI integrated FPV Goggles unit and an NUC11 computer equipped with a 4-core Intel i5-1135G7 CPU running Ubuntu 20.04. The onboard DJI O4 air unit transmits real-time video to the FPV goggles through the DJI OcuSync protocol. The goggles are connected to the DJI Fly application via a USB interface, and the video stream is wirelessly mirrored to the NUC over Wi-Fi, serving as the video input source for the system. The video transmission latency at the ground station comprises the transmission delay, DJI Fly screen casting latency, and the frame acquisition delay from the ffmpeg-based stream input. Measurements indicate a stable end-to-end latency of approximately 480 ms, which is compensated for in the subsequent visual guidance controller. In addition, the perception module is equipped with a downward-facing optical flow sensor, the MTF-01, which integrates the PMW3901 optical flow sensor and the VL53L1X laser sensor. The configuration of the avionics system and onboard sensors is illustrated in Fig. 3.

This configuration enables accurate altitude estimation and supports the vehicle's motion state localization by calculating optical flow from consecutive frames. The actuator comprises two servos and a BLDC motor, which is driven via a bidirectional DShot electronic speed controller (ESC). All actuators are controlled through PWM signals. To minimize overall weight, all avionics components have undergone decasing and weight reduction modifications.

IV. MATH MODELING

The kinematic and dynamic modeling of FWAVs is inherently complex due to the aerodynamic coupling and the nonlinear, time-varying characteristics inherent in the system. Given these challenges, we employed the cycle averaging method to simplify the model of the Sparrowhawk, deriving the unified state-space representation of its corresponding kinematic and dynamic equations.

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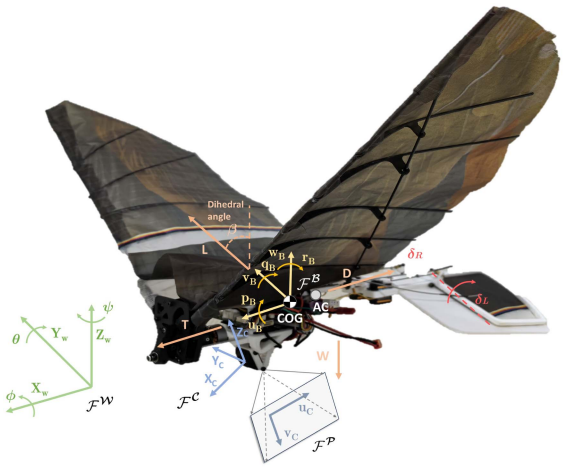


Fig. 4. Schematic of the coordinate frame and the force diagram.

The coordinate transformation involves four main frames: the world frame $\mathcal{F}^{\mathcal{W}}$, the body frame $\mathcal{F}^{\mathcal{B}}$, the camera frame $\mathcal{F}^{\mathcal{C}}$, and the pixel frame $\mathcal{F}^{\mathcal{P}}$, as shown in Fig. 4. The IMU frame is assumed to coincide with the body frame.

The position and attitude defined in the ENU world frame $\mathcal{F}^{\mathcal{W}}$ can be represented as

$$\boldsymbol{r}(t) = [x(t), y(t), z(t)]^T, \boldsymbol{\eta}(t) = [\phi(t), \theta(t), \psi(t)]^T. \quad (1)$$

The translational and rotational velocities defined in the body frame $\mathcal{F}^{\mathcal{B}}$ can be defined as

$$\boldsymbol{v}(t) = [u(t), v(t), w(t)]^T, \boldsymbol{\omega}(t) = [p(t), q(t), r(t)]^T. \quad (2)$$

Considering the coordinate transformation from the body frame to the world frame, the kinematic equation can be expressed in the following form:

$$\begin{aligned} \dot{\boldsymbol{r}}(t) &= \boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}}(\boldsymbol{\eta}(t))\, \boldsymbol{v}(t) \\ \dot{\boldsymbol{\eta}}(t) &= \boldsymbol{T}(\boldsymbol{\eta}(t))\, \boldsymbol{\omega}(t). \end{aligned} \quad (3)$$

The rotation matrix $\boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}} \in \mathrm{SO}(3)$ and the Euler angle rate transformation matrix $\boldsymbol{T} \in \mathrm{SO}(3)$ can be computed as follows, where $c_\phi \triangleq \cos\phi$ and $s_\phi \triangleq \sin\phi$

$$\boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}}(\boldsymbol{\eta}(t)) = \begin{bmatrix} c_\psi c_\theta & -c_\phi s_\psi + s_\phi s_\theta c_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ s_\psi c_\theta & c_\phi c_\psi + s_\phi s_\theta s_\psi & -s_\phi c_\psi + c_\phi s_\theta s_\psi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \quad (4)$$

$$\boldsymbol{T}(\boldsymbol{\eta}(t)) = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix}. \quad (5)$$

The forces and moments along the three axes can be simply expressed as $\boldsymbol{F} = [f_x, f_y, f_z]^T$ and $\boldsymbol{M} = [l, m, n]^T$. The dynamic equations can be derived through the Newton-Euler formula

$$\dot{\boldsymbol{v}}(t) = \begin{pmatrix} rv - qw \\ pw - ru \\ qu - pv \end{pmatrix} + \frac{1}{\mathrm{m}} \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} \quad (6)$$

$$\dot{\boldsymbol{\omega}}(t) = \boldsymbol{J}^{-1} \left[\begin{pmatrix} J_{xz} pq + (J_y - J_z)\, qr \\ J_{xz}\left(r^2 - p^2\right) + (J_z - J_x)\, pr \\ (J_x - J_y)\, pq - J_{xz} qr \end{pmatrix} + \begin{pmatrix} l \\ m \\ n \end{pmatrix} \right] \quad (7)$$

where $\boldsymbol{J}$ is the moment of inertia matrix. The diagonal elements of this matrix $J_{x,y,z}$ represent the moments of inertia, while the off-diagonal J_{xz} elements represent the products of inertia. Considering that Sparrowhawk is symmetric about the XZ-plane, the moment of inertia matrix can be formulated as

$$\boldsymbol{J} \triangleq \begin{bmatrix} J_x & 0 & -J_{xz} \\ 0 & J_y & 0 \\ -J_{xz} & 0 & J_z \end{bmatrix}. \quad (8)$$

The inverse of J can be calculated by

$$\boldsymbol{J}^{-1} = \frac{\mathrm{adj}(\boldsymbol{J})}{\det(\boldsymbol{J})} = \begin{bmatrix} J_z \Gamma^{-1} & 0 & J_{xz} \Gamma^{-1} \\ 0 & {J_y}^{-1} & 0 \\ J_{xz} \Gamma^{-1} & 0 & J_x \Gamma^{-1} \end{bmatrix} \quad (9)$$

where $\Gamma \triangleq J_x J_z - J_{xz}^2$. Hence, the state variables are denoted as $\boldsymbol{\mathcal{X}}(t) = [\boldsymbol{r}(t), \boldsymbol{\eta}(t), \boldsymbol{v}(t), \boldsymbol{\omega}(t)]^{\mathrm{T}}$. Therefore, the kinematic and dynamic state-space equations can be integrated into a unified matrix form

$$\dot{\boldsymbol{\mathcal{X}}}(t) = \begin{bmatrix} \dot{\boldsymbol{r}}(t) \\ \dot{\boldsymbol{\eta}}(t) \\ \dot{\boldsymbol{v}}(t) \\ \dot{\boldsymbol{\omega}}(t) \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_{\mathcal{B}}^{\mathcal{W}}(\boldsymbol{\eta}(t))\, \boldsymbol{v}(t) \\ \boldsymbol{T}(\boldsymbol{\eta}(t))\, \boldsymbol{\omega}(t) \\ \boldsymbol{v}(t) \times \boldsymbol{\omega}(t) \\ \boldsymbol{J}^{-1} \boldsymbol{\omega}(t)^{\wedge} \boldsymbol{J} \boldsymbol{\omega}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \boldsymbol{F} m^{-1} \\ \boldsymbol{J}^{-1} \boldsymbol{M} \end{bmatrix}. \quad (10)$$

The $\omega(t)^{\wedge}$ represents a Skey-symmetric matrix, which can be formally defined as

$$\omega(t)^{\wedge} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}. \quad (11)$$

The flapping motion of the wings is a periodic oscillation influenced by time-varying aerodynamic loads, which renders the flapping-wing dynamics a nonlinear time-periodic (NLTP) system. By applying cycle averaging theory and performing linearization, the system can be approximated as a linear time-invariant (LTI) system for analysis [12]. To analyze the aforementioned aerodynamic forces $\boldsymbol{F}$ and moments $\boldsymbol{M}$ terms, which can be decomposed into lift, drag, thrust, inertial forces, and corresponding moments. In the flapping motion, only the downstroke generates the required positive lift. The total lift is defined as the vertical component of the instantaneous lift produced by both the left and right wings. The dihedral angle is defined as β. The instantaneous lift can be determined by the following equation:

$$L(t) = \frac{1}{2} \rho C_L(t) S V^2 \cos(\beta) \quad (12)$$

where $C_L(t)$ is the lift coefficient, which increases with the angle of attack α below the critical angle, and S is the wing reference area. According to the first mean value theorem for integrals, there exists a dihedral angle within the given flapping amplitude range $-20^\circ < \beta^* < 65^\circ$, the cycle-averaged form can be expressed as [37]

$$\bar{L} = \frac{\cos\beta^*}{T} \int_0^T L(t) dt. \quad (13)$$

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Similarly, the cycle-averaged drag can be derived. The drag coefficient includes both zero-lift drag and induced drag terms, and can be expressed as

$$C_D = C_{D_0} + C_{D_i} = C_{D_0} + \frac{k^*}{\pi AT}\int_0^T C_L^2(t)dt \tag{14}$$

$$\bar{D} = \frac{1}{T}\int_0^T \frac{1}{2}\rho C_D(t) S V^2 dt. \tag{15}$$

The thrust is generated by the passive deformation of the flexible wing during flapping, resembling the fore–aft flapping motion. The $\bar{T}$ is the cycle-averaged thrust, which is responsible for counteracting the drag. The subscript v denotes the forces and moments generated by the V-wing. The cycle-averaged moment can be expressed in a generalized form as

$$\boldsymbol{M} = \frac{1}{T}\int_0^T \boldsymbol{r}(t) \times \boldsymbol{F}(t) dt. \tag{16}$$

Therefore, the forces and moments generated by the flapping wings and the V-tail can be formulated as

$$\boldsymbol{F} = \begin{bmatrix} \bar{T} + \bar{T}_v - \bar{D} - \bar{D}_v \\ 0 \\ \bar{L} + \bar{L}_v - mg \end{bmatrix}, \boldsymbol{M} = \begin{bmatrix} \bar{M}_{\text{roll}} + \bar{M}_{\text{roll},v} \\ \bar{M}_{\text{pitch}} + \bar{M}_{\text{pitch},v} \\ \bar{M}_{\text{yaw},v} \end{bmatrix}. \tag{17}$$

V. Vision-Guided Controller Design

A. Gates Detection Algorithm

In narrow gap detection at the visual front-end, accurately identifying circular gates during high-speed maneuvers with periodic oscillations poses significant challenges. The algorithm should be highly efficient and real-time capable, given the extremely limited response time for narrow gap traversal under aggressive flight conditions. First, in the absence of any prior knowledge of the gaps, such as QR codes landmarks [28] or pretrained models [22], [26], the gate detection algorithm must function without external localization assistance. Second, as the perceived radius of the gate varies markedly with distance and perspective, the detection process should be adaptive to scale changes. To meet these requirements, we propose an efficient real-time gate detection framework based on the Hough Circle Transform, enhanced with a confidence evaluation module that selects the most geometrically consistent candidates. To ensure temporal efficiency and low memory consumption while improving robustness against false detections and outliers, we apply a frame buffer combined with a sliding window mechanism to stabilize the estimates of the gate's center and radius before forwarding to the back-end controller.

The pseudocode of the visual front-end detection pipeline is presented in Algorithm 1. The framework begins by initializing the ShapeManager $\mathcal{S}$ and FrameButler $\mathcal{B}$, which are responsible for managing shape constraints and video frame buffering, respectively. For each incoming frame from the input video stream $\mathcal{V}$, the pipeline verifies whether the current target is a circular gate and invokes the comparison method to evaluate detection accuracy. Subsequently, each frame undergoes a sequence of preprocessing steps, including downsampling, grayscale, Gaussian filtering, morphological operations, and Canny edge detection. Circle candidates are extracted using the Hough Circle Transform with the following parameters

Algorithm 1: GateDetectionWithEvaluation.

Notation: Video Source $\mathcal{V}$, ShapeManager $\mathcal{S}$, FrameButler $\mathcal{B}$, Evaluator $\mathcal{E}$, Detection Result R, Circle Candidates C, Edge Image E
Input: $\mathcal{V}$, $\mathcal{S}$
Output: R
Initialize $\mathcal{S} \leftarrow$ ShapeManager($shape_id, method$);
Initialize $\mathcal{B} \leftarrow$ FrameButler($period, capacity$);
while $f_{raw} \leftarrow NextFrame(\mathcal{V})$ **do**
  if $\mathcal{S}.shape() =$ `Circle` **and** $\mathcal{S}.method() =$ `Comparison` **then**
    $f_{\text{resized}} \leftarrow$ Downsample(f_{raw});
    $f_{\text{gray}} \leftarrow$ ToGrayscale(f_{resized});
    $f_{\text{blur}} \leftarrow$ GaussianBlur(f_{gray});
    $f_{\text{morph}} \leftarrow$ MorphOpen(f_{blur});
    $E \leftarrow$ Canny(f_{morph});
    $C \leftarrow$ HoughCircles(E, $param1$, $param2$, $minRadius$, $maxRadius$);
    if $C \neq \emptyset$ **then**
      $\mathcal{E} \leftarrow$ Evaluator(E, C, $\mathcal{S}$);
      $is_rejected \leftarrow \mathcal{E}$.evaluate($fitDegree, minFitDegree$);
      $\mathcal{E}$.param2_update($param2$);
      $\mathcal{B}$.update($\mathcal{E}, is_rejected$);
      if $is_rejected$ **then**
        $\mathcal{E}_{\text{last}} \leftarrow \mathcal{B}$.getLastFrame();
        $R \leftarrow (\mathcal{E}_{\text{last}}$.center, $\mathcal{E}_{\text{last}}$.radius[0]);
      else
        $R \leftarrow (\mathcal{E}$.center, $\mathcal{E}$.radius[0]);
    else
      $param2 \leftarrow 55$;
      $\mathcal{E}_{\text{last}} \leftarrow \mathcal{B}$.getLastFrame();
      $R \leftarrow (\mathcal{E}_{\text{last}}$.center, $\mathcal{E}_{\text{last}}$.radius[0]);

param1 $= 30$, param2 $= 55$. The Evaluator class $\mathcal{E}$ assesses the geometric consistency between detected circles and the region of interest (ROI) in the target image. Specifically, each candidate circle is drawn on a zero-valued background and bitwise ANDed with the binary mask of the ROI. The number of overlapping pixels is normalized by the perimeter of the candidate circle to obtain a matching score, where a higher score indicates better spatial correspondence with the ROI. Only candidates satisfying the fit degree thresholds fitDegree $= 0.8$, minFitDegree $= 0.5$ and fall within the top matches maxFitNum $= 3$ are retained for further processing. Given that the accumulator threshold param2 regulates detection strictness and exhibits scale sensitivity, the param2_update() dynamically adjusts this threshold in proportion to the detected radius, thereby enhancing robustness under high-speed flight conditions. However, per-frame evaluation is computationally expensive and incurs high memory overhead. To ensure real-time performance and achieve low-latency frame processing, a sliding window mechanism is employed. Each frame is selected at every interval of period $= 3$ frames and the evaluated results are stored in a buffer with a size capacity $= 7$, where older candidates are automatically discarded once the buffer limit is exceeded. If valid circular candidates are identified in the current frame and the fitting degree satisfies the predefined threshold, the corresponding center and radius are adopted as the final detection result. If no candidates are available, the parameter param2 is reinitialized. In all other cases, the detection result from the previous frame is retained to ensure temporal stability.

B. Controller Design

The framework pipeline and control block diagram of the vision-based gap traversal control system is illustrated in Fig. 5, including the perception, position, and control modules. In the perception module, front-view images with a resolution

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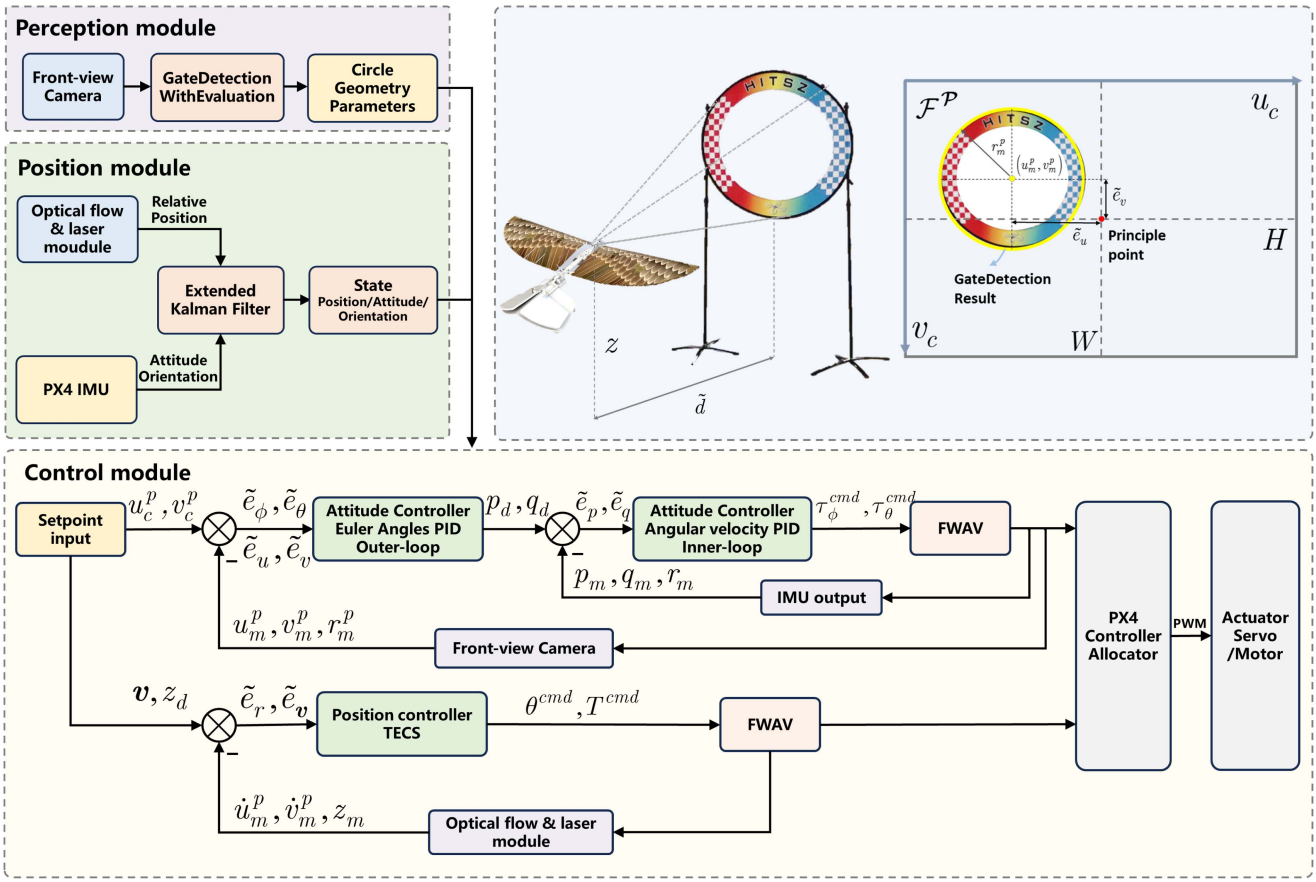


Fig. 5. Framework pipeline and control block diagram of the vision-based gap traversal controller.

of 640 × 480 captured by the onboard DJI O4 camera are compressed from 1920 × 1080 60 Hz input via the ffmpeg tool. These images are then processed using Algorithm 1 to extract the geometric features of the circular gate, where (u_m^p, v_m^p) denotes the center and r_m^p represents the radius of the detected circle. The principal point in the image pixel coordinate is defined as (u_c^p, v_c^p). The position module estimates the 6-DoF state of the robot by fusing optical flow-based visual odometry with IMU measurements via an Extended Kalman Filter (EKF). Optical flow provides relative position measurement through pixel displacement across image frames, while altitude is obtained from laser ranging. This enables the system to transition into offboard control mode for further guidance. To achieve gate traversal control, the objective is to align the detected gate center with the principal point, with the desired distance approaching zero. Consequently, the normalized pixel coordinate error and normalized distance error, which is adjusted to eliminate the influence of image resolution, can be expressed as follows:

$$\tilde{e}_u = \frac{u_m^p - u_c^p}{W}, \tilde{e}_v = \frac{v_m^p - v_c^p}{H}, \tilde{\boldsymbol{e}}_d = 1 - \sqrt{\frac{\pi r_m^{p2}}{W \times H}} \quad (18)$$

where W and H represent the width and height of the image. For vision-guided servo control based on ground station remote feedback, the total time delay primarily consists of two components: the fixed inherent end-to-end vision transmission delay $\Delta t_v = 480$ ms, and the dynamic Round Trip Time delay, Δt_c, attributable to the data link transmission and reception. The Δt_v is determined by measuring the timestamp difference between the original video stream, while the Δt_c is continuously acquired from the MAVROS topic. To address this issue, we employ a feedforward compensation strategy based on the pinhole camera model and the kinematic model derived above to predict the normalized pixel error after $\Delta t = \Delta t_v + \Delta t_c$, before entering the cascaded PID controller, thereby mitigating the effects of the delay. The normalized pixel error after feedforward compensation can be expressed as follows:

$$\tilde{\boldsymbol{e}}_{\boldsymbol{u},\boldsymbol{v}}(t + \Delta t) \approx \tilde{\boldsymbol{e}}_{\boldsymbol{u},\boldsymbol{v}}(t) - \boldsymbol{J}_\pi(\boldsymbol{P}_c(t))\omega_c(t)^\wedge \boldsymbol{P}_c(t)\Delta t \quad (19)$$

$\boldsymbol{J}_\pi$ represents the projection Jacobian matrix, which maps the velocity vector in the body frame to the velocity on the pixel plane. $\omega_c(t)^\wedge$ is the skew-symmetric matrix of angular velocity, and $P_c(t)$ is the position of the target in the camera frame. $\boldsymbol{J}_\pi$ can be derived from the camera model $Z(t)\boldsymbol{p}(t) = \boldsymbol{K}\boldsymbol{P}_c(t)$ as follows:

$$\boldsymbol{J}_p(\boldsymbol{P}_c(t)) = \begin{bmatrix} f_x Z^{-1} & 0 & -f_x X_c Z^{-2} \\ 0 & f_y Z^{-1} & -f_y Y_c Z^{-2} \end{bmatrix}. \quad (20)$$

As revealed by the state-space equations, pixel coordinate errors cannot be directly mapped to effective control inputs. Hence, the normalized roll and pitch angle errors are calculated from the aforementioned error transformation to serve as effective inputs

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(a) (b)

Fig. 6. Snapshots illustration of outdoor flight experiments. (a) Aggressive turning maneuver. (b) Autonomous flight with altitude holding.

TABLE II
COMPARATIVE RESULTS OF VISION-BASED DETECTION ALGORITHMS

Scenario	Metric	Ours	Hough circle	YOLOv 11
Dataset 1 cluster environments	Average FPS (fps)	**110.68**	55.92	13.06
	Total process time (s)	**4.23**	8.37	43.68
	CPE RMSE (pixel)	**9.81**	102.60	130.43
	Pre-training required	**No**	**No**	Yes
Dataset 2 clean dynamic	Average FPS (fps)	**142.60**	130.51	10.71
	Total process time (s)	**1.81**	1.98	28.75
	CPE RMSE (pixel)	**21.41**	27.92	83.38
	Pre-training required	**No**	**No**	Yes

for the attitude control

$$\tilde{e}_\phi = \arctan \frac{\tilde{e}_u}{\tilde{e}_d}, \tilde{e}_\theta = \arctan \frac{\tilde{e}_v}{\tilde{e}_d}. \tag{21}$$

The attitude control framework adopts a cascaded control architecture, where an outer-loop controller governs the attitude angles and an inner-loop controller regulates the angular rates. Specifically, the outer-loop PID controller computes the desired angular velocity commands based on roll and pitch errors derived from (21). These commands are compared with the real-time angular velocity measurements from the IMU to generate rate errors, which are subsequently processed by the inner-loop PID controller to output the required control torques. The longitudinal position control adopts a hierarchical TECS-based architecture, where thrust regulates total mechanical energy and pitch angle modulates the energy distribution between kinetic and potential components. Altitude and velocity errors are transformed into desired energy rate commands, with feedback from the optical flow sensor and laser rangefinder. The resulting pitch and thrust, and required control torque commands are forwarded to the control allocator and ultimately translated into actuator signals via PWM.

VI. EXPERIMENTS

To validate the aggressive maneuvering and altitude-holding capabilities of the Sparrowhawk, we conducted real-world flight and altitude-hold control experiments in an outdoor playground. Furthermore, to verify the effectiveness of the proposed vision-based guidance and control method, we carried out gap traversal real-world experiments in a confined indoor environment of approximately 20 m $\times$ 8 m.

A. Outdoor Maneuvering and Altitude-Holding Flight

To examine the practical performance of the proposed preliminary overall design, we carried out low-altitude outdoor flight tests, as shown in Fig. 6. During aggressive turning maneuvers, the robot demonstrated agile fight performance with a turning radius below 1 m. Furthermore, to validate the effectiveness of the TECS altitude-hold controller utilizing optical flow module ranging as feedback, the robot was commanded to switch to Altitude flight mode. Fig. 7 illustrates the altitude variation over a selected segment of the altitude-hold flight. The target altitude was set to 4.5 m. The raw altitude was obtained by the flight controller barometer, while the fused altitude was derived using an EKF-based estimator integrating both barometric and optical flow ranging measurements. With a tolerance band of $\pm$0.5 m defined, 81.4% of the fused altitude measurements fell within this range, exhibiting a standard deviation of 0.165 m and a mean deviation of 0.345 m.

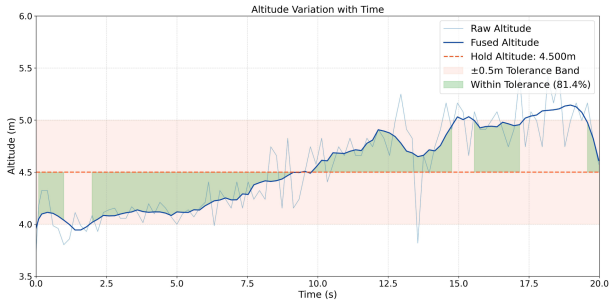


Fig. 7. Altitude variation over time during altitude-hold flight.

B. Gate Detection Algorithm Validation and Comparison

Prior to the gate traversal experiments, we validated the effectiveness and advancement of the proposed visual gate detection algorithm using a real-world circular gate video dataset and by comparison with two benchmark methods. Video datasets were recorded in both cluttered and clean dynamic environments, capturing the gate traversal process from distant to close-up perspectives. A comparative study was subsequently conducted against the unmodified Hough circle detection algorithm, serving as the traditional baseline, and the deep learning–based YOLOv11 algorithm, representing the state-of-the-art baseline. The unmodified Hough circle detector was applied with downsampling, filtering, and edge detection, employing the same parameters as our improved method. The YOLOv11 was trained on a self-constructed dataset of 200 labeled circular-gate images, captured from multiple viewpoints. To obtain the optimal model, training was performed using only a single class corresponding to the circular gate. For a fair comparison, all algorithms were executed under identical conditions on the NUC 11 computer used in our experiments, without GPU acceleration. The temporal results obtained from testing on the two datasets are shown in Fig. 8. Quantitative analysis metrics were employed to evaluate both the accuracy and efficiency of the detection algorithms. We define the circle center position error (CPE) jitter root mean square error (RMSE) to evaluate detection precision. CPE is defined as the instantaneous L_2 norm between the estimated circle center and the ground truth, as expressed by $\text{CPE}_t = ||\hat{C}_t - C_{GT}||_2$. The RMSE of the CPE sequence is subsequently employed to measure the detection's overall stability and jitter across the entire dataset

$$\text{CPE RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N} \text{CPE}_t^2}. \tag{22}$$

Furthermore, average frame rate (FPS) and total processing time were utilized to assess the detection speed, and the requirement for pre-training was used as an indicator of application

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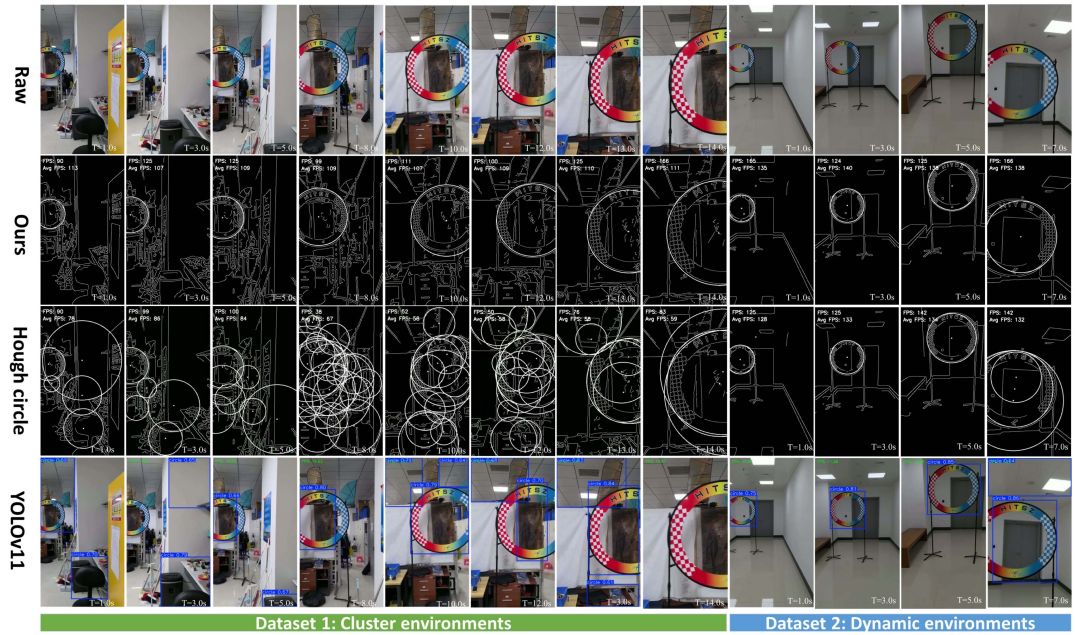


Fig. 8. Sequence comparison results of the three circular gate detection results on two datasets. Columns represent sequential video frames over time, and rows correspond to the original frame, proposed method, original Hough transform, and YOLOv11 algorithm results.

complexity. These quantitative metrics for the algorithms are summarized in Table II. Experimental results show that the proposed detection algorithm achieves superior performance in terms of average frame rate, detection robustness, and recognition accuracy under both cluttered and dynamic environments. In the cluttered environment, the conventional Hough detector produced multiple false positives, and YOLOv11 exhibited similar shortcomings. Moreover, YOLOv11 failed to accurately estimate the geometric center of the gate when only partial circular features were present, and without GPU acceleration achieved an average frame rate of only about 10 fps. In the clean dynamic dataset, all three algorithms initially provided relatively stable detections. However, as the gate's pixel size increased, the conventional Hough detector and YOLOv11 failed to accurately and consistently track and recognize the circular gate.

C. Vision-Guided Indoor Narrow Gap Traversal

Drawing inspiration from the agile maneuverability and obstacle avoidance capabilities exhibited by sparrowhawks navigating through constrained environments, we conducted multiple real-world indoor experiments involving narrow-gap traversal under varying lighting conditions to evaluate the performance of the proposed system. The diameter of the circular gate was set to 80 cm, while the Sparrowhawk's wingspan is 60 cm, leaving only 10 cm of horizontal and vertical safety clearance, requiring precise whole-body control to flight through the center of the gate. The dimensional comparison between the Sparrowhawk and the circular gate is illustrated in Fig. 9.

The gate traversal experiments are conducted as follows. After being hand-launched, the robot autonomously performs gate detection, levels its flight under servo control, and flies through the circular gate. Localization data is acquired using an onboard optical flow module, which allows the robot to autonomously switch to offboard control mode upon detecting the circular gate. Angular velocity commands, derived from a cascaded attitude-loop PID controller, and thrust commands generated by

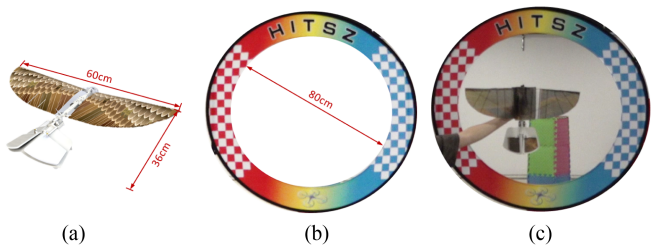


Fig. 9. (a) Sparrowhawk dimensions. (b) Circular gate dimensions. (c) Geometric size comparison between the Sparrowhawk and the gate.

the TECS-based position loop will be transmitted to the PX4 control allocator through MAVROS. The allocator generates DShot commands for the ESCs to maintain altitude and PWM signals for the servos to control attitude. After takeoff, the vision-guided controller effectively aligns the camera's principal point with the center of the circular gate, guiding the robot to traverse through the gate. We successfully performed gate traversal under varying lighting conditions and carried out repeated experiments to enable comparative analysis. Fig. 10 shows frontal and side views of the robot successfully traversing the circular gate under both daytime and low-light conditions, along with the corresponding FPV sequence. Flight trajectory snapshots under daytime normal illumination and nighttime low-light conditions are shown in Fig. 10(a) and (c). The corresponding FPV camera images and the edge-detection outputs of the gate detection algorithm are displayed in the first and second rows of Fig. 10(b) and (d), respectively. To make a comprehensively evaluate the generalizability and scalability of the narrow gap traversal task, multiple groups of gate traversal success rate experiments were conducted under varying lighting conditions. For baseline comparison, a professional human pilot, the unmodified Hough circle algorithm, and the YOLOv11 detector, both of which employ the same cascaded PID control scheme as our proposed system, were subjected to the same repeated experiments under

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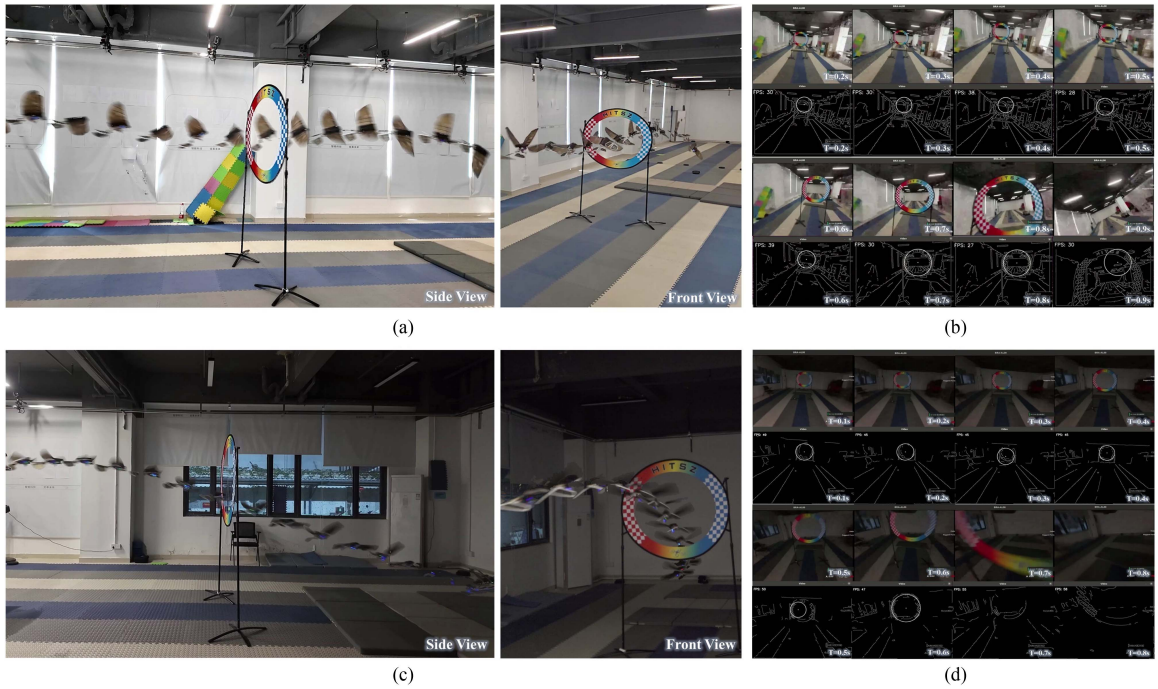


Fig. 10. Snapshot sequence of Sparrowhawk performing narrow circular gate traversal under varying illumination conditions. (a) Agile narrow-gap gate traversal flight trajectory under normal illumination. (b) Sequence diagram of gap traversal from FPV perspective under normal illumination. (c) Agile narrow-gap gate traversal flight trajectory under low-light conditions. (d) Sequence diagram of gap traversal from FPV perspective under low-light conditions.

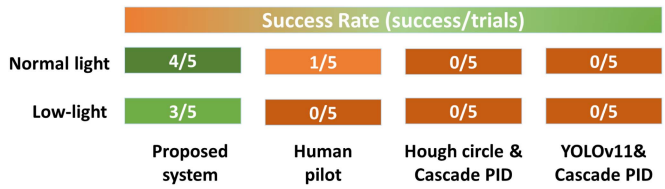


Fig. 11. Comparison of gate-traversal success rates across different lighting conditions and frameworks.

identical conditions. Based on repeated flight tests, with at least five trials per set, the proposed system achieved an 80% success rate under normal lighting and a 60% success rate in low-light conditions. Although low-light conditions slightly degrade the performance of the vision-based front-end, the system maintains a robust success rate. In comparison, the human pilot achieved a 10% success rate under normal lighting and failed entirely under low-light conditions. The low-light scenario posed a significant challenge for the human pilot, who struggled to complete the complex maneuver sequence within the limited reaction time, using only a third-person perspective. Meanwhile, frameworks relying on the unmodified Hough circle detector and YOLOv11 failed to provide accurate gate-center geometry, leading to erroneous visual-servo control. The success rate results are presented in Fig. 11.

To illustrate the traversal process in detail, representative data from successful gate traversal experiments under normal illumination conditions are presented. The quantitative results of these maneuvers are shown in Figs. 12 and 13. The traversal maneuver flight process was completed within 1 s. During the traversal, the oscillation amplitudes of the roll and pitch angles are 1.368 rad and 1.097 rad, respectively, with maximum deviations of 2.251 rad and 2.176 rad. The oscillation amplitudes of the roll and pitch

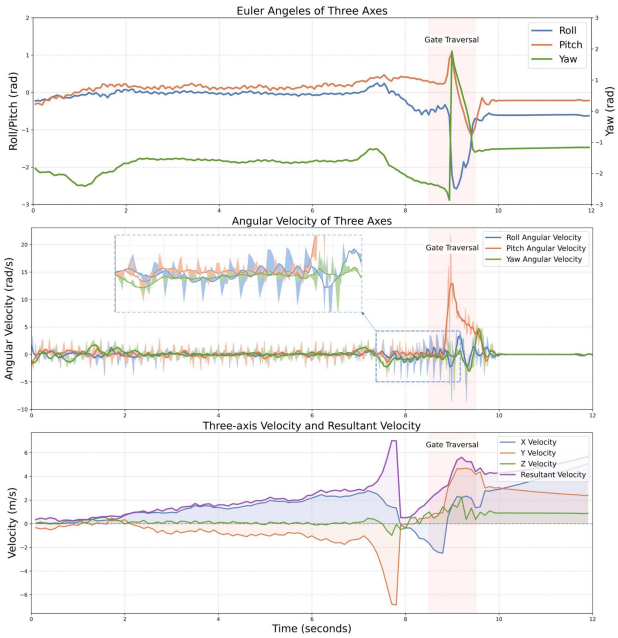


Fig. 12. Temporal evolution of Euler angles, angular velocities, and three-dimensional resultant translational velocities.

angular velocities are 3.508 rad/s and 12.949 rad/s, showing a strong correlation with the variations in the normalized roll and pitch angular velocity setpoints generated by the cascaded PID controller. This corelation is also reflected in the sudden transients in the pitch/roll angles and angular velocities. Specifically, after the robot levels out from takeoff, the visual servo controller outputs pitch/roll angular velocity setpoints based on the position of the principal point relative to the identified gate

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TABLE III
COMPARATIVE ANALYSIS OF VALIDATION RESULTS WITH OTHER FLAPPING-WING PLATFORMS

Comparison indicators	Evaluation Metric	Ours	Rocbofalcon [1], [38]	Dove [6], [36]
Trim values (Maintain 6 m/s steady flight)	Flapping frequency (Hz)	7.5	3.0	8.0
	Flapping amplitude (Deg)	85	85	73
	Angle of attack (Deg)	15	12	15
System identification and eigenvalues of the cycle-averaged LTI system	Eigenvalues	$\lambda_1 = -0.019, \lambda_{2,3} = -0.505 \pm 2.563j$	$\lambda_1 = 0, \lambda_{2,3} = -1.075 \pm 1.656j$	NaN
	Natural frequency and Damping ratio	$\omega_n \approx 2.612$rad/s, $\zeta \approx 0.193$	$\omega_n \approx 1.973$rad/s, $\zeta \approx 0.545$	
Flight performance	Total weight (g)	245	600	220
	Wingspan (cm)	60	120	50
	Flapping frequency (Hz)	0-8	0-5.5	0-12
	Flight speed (m/s)	4-10	6-12	8-12
	Maximum endurance (min)	20	11	30

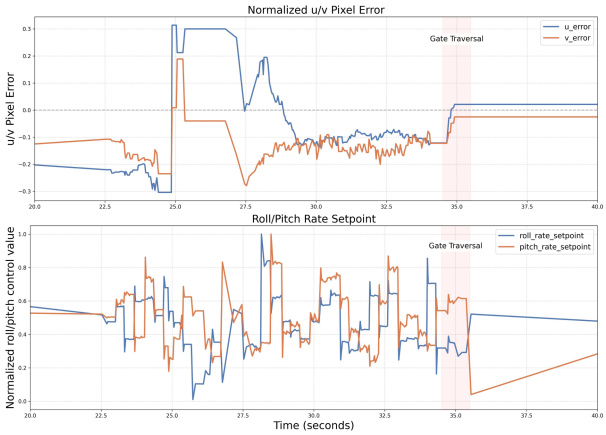


Fig. 13. Temporal evolution of normalized pixel errors in u and v axes, and control setpoint of roll and yaw angular rates.

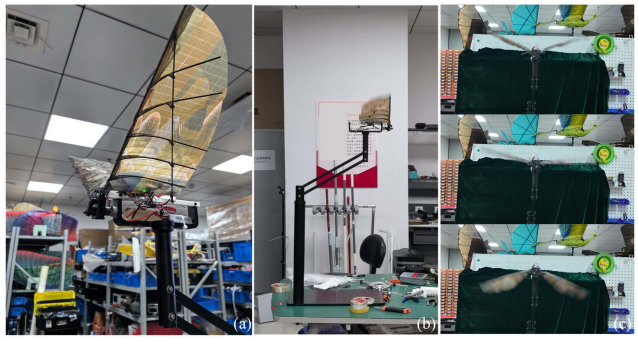


Fig. 14. (a) Setup of the static flapping prototype. (b) Sideview of the static flapping experiment. (c) Snapshot of the downstroke motion.

center. This manifests as an increase in pitch for rapid climbing to the gate's height, and a large roll maneuver to execute the necessary left turn through the center. The peak resultant velocity during gate traversal reached 5.6 m/s, while the normalized mean absolute tracking errors along the u and v axis in the pixel coordinate system are 4.21% and 6.15%, respectively. The satisfactory results of the gate traversal control validate the effectiveness of the proposed vision-based guidance and control method.

VII. DISCUSSION

A. Comparative Validation With Other Platforms

To provide further validation for our proposed design and model, we have conducted a detailed numerical comparison with state-of-the-art flapping-wing platforms reported in the related literature, examining their trim values, eigenvalues of the linearized model, and flight performance parameters. The detailed comparison results are presented in Table III. Specifically, trim values represent the steady-state equilibrium of the aircraft under a specific flight condition. For our validation, a trim state is defined by the flapping frequency, flapping amplitude, and angle of attack required to maintain a steady flight speed of 6 m/s. In Section IV, the nonlinear and quasi-steady aerodynamic NLTP system at low Reynolds numbers is reformulated as an LTI system through cycle-averaging and linearization, enabling stability assessment via the eigenvalues of the system matrix [39], [40]. For comparison, the lateral dynamics identification parameters reported in [1] are adopted as a benchmark. Based on periodic averaging and small-disturbance theories [41], along with the previously derived state-space model, system identification is performed using experimental data from gap traversal flight tests. The eigenvalues of the linearized model, together with the computed natural frequencies and damping ratios, confirm that the system exhibits lateral dynamic stability. In addition, flight performance parameters are presented for comparative analysis.

B. Static Flapping Experiments and Durability Validation

To evaluate the durability and reliability of the Sparrowhawk, we conducted continuous static flapping experiments to assess the durability of the transmission system, wings, and airframe mechanical structure, as illustrated in Fig. 14. The prototype was mounted on a torsional spring support bracket that allows a single degree of freedom motion along the z-axis, and was operated continuously for more than one hour at throttle levels exceeding 50%. Results from both indoor/outdoor flight tests and static flapping experiments demonstrate that the transmission system and wings of the prototype can withstand over 100 h of continuous flapping. Moreover, the prototype can sustain repeatable performance for over 50 consecutive flights without severe landing impacts.

VIII. CONCLUSION

Emulating avian obstacle avoidance maneuvers could significantly enhance the survival, operational safety, and environmental adaptability of FWAVs in industrial scenarios. Achieving fully autonomous, safety-assured navigation necessitates improvements in flight performance, state estimation accuracy, and precise perception-driven control. To address the challenges of high-load flight, onboard sensing, and servo control encountered by FWAVs, this article introduces Sparrowhawk, detailing its mechanical design, sensor integration, modeling, and proposes

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a vision-guided real-time closed-loop control framework. The superiority of the proposed vision-based detection algorithm is validated through benchmark comparison experiments. Indoor and outdoor flight tests demonstrate that the system can perform tight-radius banked turns, altitude-holding flight, and autonomous circular gap traversal with only 10 cm safety margins at a flight speed of 5.6 m/s. In addition, multiple repeated experiments under varying lighting conditions confirm the robustness and scalability of the proposed system. Future work will focus on enhancing the robot's onboard computing capabilities and training more accurate dynamic models via learning-based methods to support the deployment of advanced control strategies.

REFERENCES

[1] A. Chen, B. Song, Z. Wang, D. Xue, and K. Liu, "A novel actuation strategy for an agile bioinspired FWAV performing a morphing-coupled wingbeat pattern," *IEEE Trans. Robot.*, vol. 39, no. 1, pp. 452–469, Feb. 2023.

[2] C. Wu, Y. Xiao, J. Zhao, J. Mou, F. Cui, and W. Liu, "A multi-modal tailless flapping-wing robot capable of flying, crawling, self-righting and horizontal take-off," *IEEE Robot. Autom. Lett.*, vol. 9, no. 5, pp. 4734–4741, May 2024.

[3] S. Kim, Y.-H. Hsiao, Z. Ren, J. Huang, and Y. Chen, "Acrobatics at the insect scale: A durable, precise, and agile micro–aerial robot," *Sci. Robot.*, vol. 10, no. 98, Jan. 2025, Art. no. eadp4256.

[4] R. Zufferey, J. Tormo-Barbero, D. Feliu-Talegón, S. R. Nekoo, J. Á. Acosta, and A. Ollero, "How ornithopters can perch autonomously on a branch," *Nature Commun.*, vol. 13, no. 1, Dec. 2022, Art. no. 7713.

[5] F. J. Maldonado, J. A. Acosta, J. Tormo-Barbero, P. Grau, M. M. Guzman, and A. Ollero, "Adaptive nonlinear control for perching of a bioinspired ornithopter," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Oct. 2020, pp. 1385–1390.

[6] W. Yang, L. Wang, and B. Song, "Dove: A biomimetic flapping-wing micro air vehicle," *Int. J. Micro Air Veh.*, vol. 10, no. 1, pp. 70–84, Mar. 2018.

[7] D. Gayango, R. Salmoral, H. Romero, J. M. Carmona, A. Suarez, and A. Ollero, "Benchmark evaluation of hybrid fixed-flapping wing aerial robot with autopilot architecture for autonomous outdoor flight operations," *IEEE Robot. Autom. Lett.*, vol. 8, no. 7, pp. 4243–4250, Jul. 2023.

[8] Q. Fu, S. Wang, J. Wang, X. Wu, and W. He, "Target detection and localization for flapping-wing robots based on biological eagle-eye vision," *IEEE/ASME Trans. Mechatronics*, vol. 30, no. 2, pp. 1590–1600, Apr. 2025.

[9] S. R. Nekoo and A. Ollero, "Closed-loop nonlinear optimal control design for flapping-wing flying robot (1.6 m wingspan) in indoor confined space: Prototyping, modeling, simulation, and experiment," *ISA Trans.*, vol. 142, pp. 635–652, Nov. 2023.

[10] K. Y. W. Scheper, M. Karasek, C. De Wagter, B. D. W. Remes, and G. C. H. E. De Croon, "First autonomous multi-room exploration with an insect-inspired flapping wing vehicle," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2018, pp. 5546–5552.

[11] M. Karásek, F. T. Muijres, C. De Wagter, B. D. W. Remes, and G. C. H. E. de Croon, "A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns," *Science*, vol. 361, no. 6407, pp. 1089–1094, Sep. 2018.

[12] S. Tijmons, G. C. H. E. de Croon, B. D. W. Remes, C. De Wagter, and M. Mulder, "Obstacle avoidance strategy using onboard stereo vision on a flapping wing MAV," *IEEE Trans. Robot.*, vol. 33, no. 4, pp. 858–874, Aug. 2017.

[13] H. V. Phan, S. Aurecianus, T. K. L. Au, T. Kang, and H. C. Park, "Towards the long-endurance flight of an insect-inspired, tailless, two-winged, flapping-wing flying robot," *IEEE Robot. Autom. Lett.*, vol. 5, no. 4, pp. 5059–5066, Oct. 2020.

[14] H. V. Phan and H. C. Park, "Mechanisms of collision recovery in flying beetles and flapping-wing robots," *Science*, vol. 370, no. 6521, pp. 1214–1219, Dec. 2020.

[15] Y.-W. Chin et al., "Efficient flapping wing drone arrests high-speed flight using post-stall soaring," *Sci. Robot.*, vol. 5, no. 44, Jul. 2020, Art. no. eaba2386.

[16] R. Zufferey et al., "Design of the high-payload flapping wing robot E-flap," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3097–3104, Apr. 2021.

[17] E. Pan, H. Xu, H. Yuan, J. Peng, and W. Xu, "HIT-Hawk and HIT-Phoenix: Two kinds of flapping-wing flying robotic birds with wingspans beyond 2 meters," *Biomimetic Intell. Robot.*, vol. 1, Jun. 2021, Art. no. 100002.

[18] X. Wu, W. He, Q. Wang, T. Meng, X. He, and Q. Fu, "A long-endurance flapping-wing robot based on mass distribution and energy consume method," *IEEE Trans. Ind. Electron.*, vol. 70, no. 8, pp. 8215–8224, Aug. 2023.

[19] R. Meng, B. Song, J. Xuan, and X. Yang, "Design and verification of a large-scaled flapping-wing aircraft named 'cloud Owl'," *Appl. Sci.*, vol. 13, no. 9, May 2023, Art. no. 5667.

[20] S. Ryu, U. Kwon, and H. J. Kim, "Autonomous flight and vision-based target tracking for a flapping-wing MAV," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Oct. 2016, pp. 5645–5650.

[21] M. Wang et al., "Unlocking aerobatic potential of quadcopters: Autonomous freestyle flight generation and execution," *Sci. Robot.*, vol. 10, 2025, Art. no. eadp9905.

[22] Y. Xie, M. Lu, R. Peng, and P. Lu, "Learning agile flights through narrow gaps with varying angles using onboard sensing," *IEEE Robot. Autom. Lett.*, vol. 8, no. 9, pp. 5424–5431, Sep. 2023.

[23] B. Penin, P. R. Giordano, and F. Chaumette, "Vision-based reactive planning for aggressive target tracking while avoiding collisions and occlusions," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3725–3732, Oct. 2018.

[24] Y. Ren et al., "Safety-assured high-speed navigation for MAVs," *Sci. Robot.*, vol. 10, 2025, Art. no. eado6187.

[25] J. Tordesillas, B. T. Lopez, M. Everett, and J. P. How, "FASTER: Fast and safe trajectory planner for navigation in unknown environments," *IEEE Trans. Robot.*, vol. 38, no. 2, pp. 922–938, Apr. 2022.

[26] J. Lin, L. Wang, F. Gao, S. Shen, and F. Zhang, "Flying through a narrow gap using neural network: An end-to-end planning and control approach," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Nov. 2019, pp. 3526–3533.

[27] S. Chen, Y. Li, Y. Lou, K. Lin, and X. Wu, "Learning real-time dynamic responsive gap-traversing policy for quadrotors with safety-aware exploration," *IEEE Trans. Intell. Veh.*, vol. 8, no. 3, pp. 2271–2284, Mar. 2023.

[28] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, and D. Scaramuzza, "Champion-level drone racing using deep reinforcement learning," *Nature*, vol. 620, no. 7976, pp. 982–987, Aug. 2023.

[29] Y. Song, A. Romero, M. Müller, V. Koltun, and D. Scaramuzza, "Reaching the limit in autonomous racing: Optimal control versus reinforcement learning," *Sci. Robot.*, vol. 8, no. 82, Sep. 2023, Art. no. eadg1462.

[30] G. C. H. E. De Croon, E. De Weerdt, C. De Wagter, B. D. W. Remes, and R. Ruijsink, "The appearance variation cue for obstacle avoidance," *IEEE Trans. Robot.*, vol. 28, no. 2, pp. 529–534, Apr. 2012.

[31] C. De Wagter, S. Tijmons, B. D. W. Remes, and G. C. H. E. de Croon, "Autonomous flight of a 20-gram flapping wing MAV with a 4-gram onboard stereo vision system," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2014, pp. 4982–4987.

[32] J. Jiang, E. Pan, W. Xu, W. Sun, and J. Ye, "FWAF-VID: A flapping-wing aggressive flight benchmark dataset for visual-inertial localization," *IEEE Robot. Autom. Lett.*, vol. 10, no. 6, pp. 6328–6335, Jun. 2025.

[33] J. P. Rodriguez-Gomez et al., "The GRIFFIN perception dataset: Bridging the gap between flapping-wing flight and robotic perception," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1066–1073, Apr. 2021.

[34] A. G. Eguiluz et al., "Why fly blind? Event-based visual guidance for ornithopter robot flight," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2021, pp. 1958–1965.

[35] J. Lee, S. Ryu, T. Kim, W. Kim, and H. J. Kim, "Learning-based path tracking control of a flapping-wing micro air vehicle," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Oct. 2018, pp. 7096–7102.

[36] C. Chen, B. Song, Q. Fu, and J. Gao, "Autonomous navigation system for flapping wing aerial vehicle based on event-trigger planner," *Chin. J. Aeronaut.*, vol. 38, May 2025, Art. no. 103576.

[37] G. Sachs, "Comparison of power requirements: Flapping vs. fixed wing vehicles," *Aerospace*, vol. 3, no. 4, Sep. 2016, Art. no. 31.

[38] A. Chen, B. Song, Z. Wang, K. Liu, D. Xue, and X. Yang, "Experimental study on the effect of increased downstroke duration for an FWAV with morphing-coupled wing flapping configuration," *J. Bionic. Eng.*, vol. 21, pp. 192–208, Oct. 2023.

[39] E. Yesilbas, B. Özgür, E. Ozen, and T. Oktay, "Simultaneous and stochastic design of piston-prop TUAV vertical tail and its autonomous system," *AEAT*, vol. 97, no. 4, pp. 395–404, Apr. 2025.

[40] N. Ünal, Y. Öz, E. A. Ünal, and T. Oktay, "Enhancing aerodynamic performance of a two-dimensional airfoil using plasma actuators," *Aerosp. Sci. Technol.*, vol. 158, Mar. 2025, Art. no. 109882.

[41] J. Grauer, E. Ulrich, J. Hubbard, D. Pines, and J. S. Humbert, "Testing and system identification of an ornithopter in longitudinal flight," *J. Aircr.*, vol. 48, no. 2, pp. 660–667, Mar. 2011.

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